Video 03 | Wavetables and Oscillators

Massive PSXL instructional series - video script

Runtime estimate	Narration length	Scenes	Format
about 5:56 minutes	841 words at 150 words per minute	13	Screen capture of the plug-in with voice-over; dark theme, interface size 125 percent

Learning goals

- Understand frames, Pos and the ten built-in wavetables.
- Use Pitch, Fine, Unison, Detune, PhaseMod and Fold.
- Import your own audio file as a wavetable.

> **NOTE.** Accuracy rule for the narrator and any generated video: use the on-screen names exactly as written in the action column. After **Load Init (Blank)** the synth is silent until routing is connected. Do not compare Massive PSXL to other products.

Scene table

Scene	On-screen action	Narration (word for word)	Visual notes and call-outs
1	Start from Init (Blank), quick setup Osc 1 > Filter 1 > Out; Filter 1 Cutoff 12000.	Start from Init Blank with the quick setup Osc 1 into Filter 1 into Out. We will keep the filter open and just listen to the oscillator. If you have just come from the second video, you will recognise this setup. If not, pause and build it; it takes about ten seconds.	Brief title card: 'Wavetables and oscillators'.
2	Drag in the wavetable display slowly left to right while holding a low note. Show the highlighted line.	A wavetable is a stack of single-cycle waveforms called frames. The 3D display draws them back to front. The bright line is the frame you hear, and Pos moves it. Between frames the sound is crossfaded, so the morph is smooth. Drag slowly while holding a low note and listen for the smooth changes. If you hear steps, move slower. The position knob in the panel does exactly the same thing as dragging in the display.	Slow pan across the display with a callout 'frame = one single-cycle wave'.
3	Use the wavetable menu and the < > buttons to step through Morph Basic, Pulse Sweep and Harmonic Tilt, sweeping Pos in each.	Ten tables are built in. Morph Basic goes from sine, through triangle and saw, to square. Pulse Sweep narrows a pulse wave, a classic for pulse width movement. Harmonic Tilt goes from dull to very bright. Try the previous and next buttons beside the wavetable menu to flick through tables quickly. They only step through the tables that are loaded, including any you import.	Overlay the position marks: sine, triangle, saw, square at 0, 0.33, 0.66, 1.0 on Morph Basic.
4	Step through Vowel Choir, Hard Sync, FM Bells; sweep Pos.	Vowel Choir sings A, E, I, O, U as you move Pos. Hard Sync sweeps a synced saw, and FM Bells goes from a soft sine to a bright, metallic strike. A tip for the choir: the vowels are baked into the waveform, so they move with the note you play. They sound most natural in the low to middle register.	Overlay A E I O U along the Pos axis for Vowel Choir.

Scene	On-screen action	Narration (word for word)	Visual notes and call-outs
5	Step through Drawbar Organ, Glass Harmonics, Digital Grit, Fold Sweep; sweep Pos.	Drawbar Organ morphs through organ registrations. Glass Harmonics has a bright peak that travels up the spectrum. Digital Grit is a harsh, random-phase texture, and Fold Sweep progressively folds a sine wave back on itself. Each of these tables is thirty-two frames long. That is plenty to make smooth sweeps, and they are band-limited, so they stay clean even high up the keyboard.	Show the table name appearing in the display corner each time.
6	Set Osc 1 to Morph Basic, Pos 0.66. Turn Pitch to +12 and back; then Fine -20 to +20.	Pitch tunes in whole semitones, up to four octaves either way. Fine tunes in cents. Together they let you build intervals and detune against the other oscillator. Intervals are easy. Set Osc 2 to plus seven, a fifth, or plus twelve for an octave. For a thicker single sound, use Fine to detune Osc 2 just a few cents away from Osc 1.	Show hover readouts in st and ct.
7	Raise Unison from 1 to 8 while holding a chord; turn Detune up and down.	Unison stacks up to eight copies of the oscillator. Detune spreads them. One voice is a single saw. Eight with about thirty cents of detune is a supersaw. The volume stays steady, because the stack is scaled automatically. Be aware that it costs CPU. As you raise detune, the stack gets wider and the beating gets faster. Small amounts feel like a warm chorus; large amounts feel like an ensemble of many oscillators. Each voice starts at a random phase, so every note sounds slightly different.	Show Unison reading 1, 3, 8. Callout: 'Eight unison x eight notes = 64 oscillators'.
8	Activate Osc 2: Routing wire Osc 2 > Filter 1. Osc 2 Pitch +7. Compare Osc 2 table choice.	Both oscillators have the same controls. Wire Osc 2 into Filter 1 as well, and give it its own table, pitch and unison. Remember that Osc 2 starts on Pulse Sweep, not Morph Basic. A different table in each oscillator is a quick way to get complex tones. A bell table on Osc 2 under a saw on Osc 1, for example, is a good starting point for keys.	Split screen: Osc 1 tab and Osc 2 tab controls.
9	Osc 2 Level to 0 (and leave it unwired in a second take). Raise Osc 1 PhaseMod to 0.3 and sweep Osc 2 Pitch.	Phase mod is FM-style. Each oscillator can phase-modulate the other. Raise PhaseMod on Osc 1 and Osc 2 becomes the modulator. Both oscillators always run, so even with Osc 2's level at zero, or with no wire at all, it still modulates. Change its pitch to change the colour. The modulator's pitch sets the ratio between the two oscillators. Whole-number ratios sound harmonic and musical; odd ratios sound like bells and clangs.	Diagram overlay: Osc 2 -> arrow -> Osc 1 phase.
10	Turn Fold up from 0 to 0.6 on Osc 1 while holding a note.	Fold pushes the oscillator into a wavefolder. Gently it adds harmonics; more, and the sound goes hollow and metallic. A little fold makes a saw more complex and vocal; a lot makes it hollow and harsh. It reacts to level, so unison and the choice of table matter. Try modulating it with an envelope for sounds that start rough and mellow out.	Waveform display shows folding if available; otherwise a graphic of a folded sine.
11	Drag a WAV file onto the wavetable display. The toast appears; the table shows 'User: filename'. Sweep Pos.	To use your own sound, drag an audio file onto the display. WAV, AIFF, FLAC or OGG all work. Massive PSXL slices it into frames of two thousand forty-eight samples, up to two hundred fifty-six frames, and the new table appears in the menu, named after your file. Single-cycle files of two thousand forty-eight samples give the cleanest results, but any audio works. Recordings of voices or instruments turn into evolving textures, and Pos becomes a way to scrub through the recording.	Show the toast 'Imported wavetable ... into Osc 1'. Show Finder window with the file being dragged.
12	Press the + button beside the wavetable menu; show Menu > Settings > Import wavetable...; show the Voice tab list of user tables.	You can also press the plus button, or use Menu, Settings, Import wavetable. The Voice tab lists your user tables. There are twenty-two user slots, and tables are saved with your presets and your project. When all twenty-two user slots are full, the last one is replaced, so keep an eye on your library. If a file cannot be read, a message tells you.	Callout on each of the three import methods in sequence.

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13	End on a sound using the imported wavetable.	That is the oscillator section. Try Unison, PhaseMod and Fold together on a table you like, and join us in video four for filters and the insert.	End card: 'Next: 04 Filters and the Insert'.

Narration only (plain script for text-to-speech or NotebookLM)

[Scene 1] Start from Init Blank with the quick setup Osc 1 into Filter 1 into Out. We will keep the filter open and just listen to the oscillator. If you have just come from the second video, you will recognise this setup. If not, pause and build it; it takes about ten seconds.

[Scene 2] A wavetable is a stack of single-cycle waveforms called frames. The 3D display draws them back to front. The bright line is the frame you hear, and Pos moves it. Between frames the sound is crossfaded, so the morph is smooth. Drag slowly while holding a low note and listen for the smooth changes. If you hear steps, move slower. The position knob in the panel does exactly the same thing as dragging in the display.

[Scene 3] Ten tables are built in. Morph Basic goes from sine, through triangle and saw, to square. Pulse Sweep narrows a pulse wave, a classic for pulse width movement. Harmonic Tilt goes from dull to very bright. Try the previous and next buttons beside the wavetable menu to flick through tables quickly. They only step through the tables that are loaded, including any you import.

[Scene 4] Vowel Choir sings A, E, I, O, U as you move Pos. Hard Sync sweeps a synced saw, and FM Bells goes from a soft sine to a bright, metallic strike. A tip for the choir: the vowels are baked into the waveform, so they move with the note you play. They sound most natural in the low to middle register.

[Scene 5] Drawbar Organ morphs through organ registrations. Glass Harmonics has a bright peak that travels up the spectrum. Digital Grit is a harsh, random-phase texture, and Fold Sweep progressively folds a sine wave back on itself. Each of these tables is thirty-two frames long. That is plenty to make smooth sweeps, and they are band-limited, so they stay clean even high up the keyboard.

[Scene 6] Pitch tunes in whole semitones, up to four octaves either way. Fine tunes in cents. Together they let you build intervals and detune against the other oscillator. Intervals are easy. Set Osc 2 to plus seven, a fifth, or plus twelve for an octave. For a thicker single sound, use Fine to detune Osc 2 just a few cents away from Osc 1.

[Scene 7] Unison stacks up to eight copies of the oscillator. Detune spreads them. One voice is a single saw. Eight with about thirty cents of detune is a supersaw. The volume stays steady, because the stack is scaled automatically. Be aware that it costs CPU. As you raise detune, the stack gets wider and the beating gets faster. Small amounts feel like a warm chorus; large amounts feel like an ensemble of many oscillators. Each voice starts at a random phase, so every note sounds slightly different.

[Scene 8] Both oscillators have the same controls. Wire Osc 2 into Filter 1 as well, and give it its own table, pitch and unison. Remember that Osc 2 starts on Pulse Sweep, not Morph Basic. A different table in each oscillator is a quick way to get complex tones. A bell table on Osc 2 under a saw on Osc 1, for example, is a good starting point for keys.

[Scene 9] Phase mod is FM-style. Each oscillator can phase-modulate the other. Raise PhaseMod on Osc 1 and Osc 2 becomes the modulator. Both oscillators always run, so even with Osc 2's level at zero, or with no wire at all, it still modulates. Change its pitch to change the colour. The modulator's pitch sets the ratio between the two oscillators. Whole-number ratios sound harmonic and musical; odd ratios sound like bells and clangs.

[Scene 10] Fold pushes the oscillator into a wavefolder. Gently it adds harmonics; more, and the sound goes hollow and metallic. A little fold makes a saw more complex and vocal; a lot makes it hollow and harsh. It reacts to level, so unison and the choice of table matter. Try modulating it with an envelope for sounds that

start rough and mellow out.

[Scene 11] To use your own sound, drag an audio file onto the display. WAV, AIFF, FLAC or OGG all work. Massive PSXL slices it into frames of two thousand forty-eight samples, up to two hundred fifty-six frames, and the new table appears in the menu, named after your file. Single-cycle files of two thousand forty-eight samples give the cleanest results, but any audio works. Recordings of voices or instruments turn into evolving textures, and Pos becomes a way to scrub through the recording.

[Scene 12] You can also press the plus button, or use Menu, Settings, Import wavetable. The Voice tab lists your user tables. There are twenty-two user slots, and tables are saved with your presets and your project. When all twenty-two user slots are full, the last one is replaced, so keep an eye on your library. If a file cannot be read, a message tells you.

[Scene 13] That is the oscillator section. Try Unison, PhaseMod and Fold together on a table you like, and join us in video four for filters and the insert.

> **TIP.** Pause for about one second between scenes and let on-screen actions finish before the next sentence starts.